

Fortsetzung Leistungen:

$$\text{Üb.: } \underline{U} = 126 \text{ V} \cdot e^{-j44^\circ}$$

$$\underline{I} = 12 \text{ A} \cdot e^{+j13^\circ}$$

$$\Rightarrow P, Q, S, \varphi, \lambda, S$$

$$\underline{S} = \underline{U} \cdot \underline{I}^*$$

$$= 126 \text{ V} \cdot e^{-j44^\circ} \cdot 12 \text{ A} \cdot e^{-j13^\circ}$$

$$\underline{S} = 1512 \text{ VA} \cdot e^{-j57^\circ}$$

$$|\underline{S}| = S = 1512 \text{ VA} \quad \text{Scheinleistung (apparent power)}$$

$$\underline{S} = P + jQ$$

$$= 1512 \text{ VA} \cdot [\cos(-57^\circ) + j \sin(-57^\circ)]$$

$$= \underbrace{823,5 \text{ VA}}_P - j \underbrace{1268 \text{ VA}}_Q$$

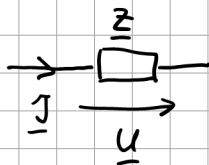
$$P = 823,5 \text{ W} \quad \text{Wirkleistung (active power)}$$

$$Q = -1268 \text{ var} \quad \text{Blindleistung (reactive power)}$$

$$\lambda = \frac{P}{S} = \cos \varphi = \cos(-57^\circ) = 0,54$$

$$P > 0 \Rightarrow \text{Verbraucher}$$

$$Q < 0 \Rightarrow \text{kapazitiv}$$

LE1

$$U = 12,4V$$

$$\varphi_u = 28^\circ$$

$$I = 1,5A$$

$$\varphi_i = -76^\circ$$

gesucht: $P, Q, S, \underline{S}, \varphi$, Charakteristik von $\underline{Z}$ (Verbraucher, Erzeuger)
(ind., kap.)

$$\underline{U} = 12,4V \cdot e^{j28^\circ}$$

$$\underline{I} = 1,5A \cdot e^{-j76^\circ}$$

$$\begin{aligned}\underline{S} &= \underline{U} \cdot \underline{I}^* = 12,4V \cdot e^{j28^\circ} \cdot 1,5A \cdot e^{+j76^\circ} \\ &= 18,6VA \cdot e^{j104^\circ}\end{aligned}$$

$$|\underline{S}| = S = 18,6VA$$

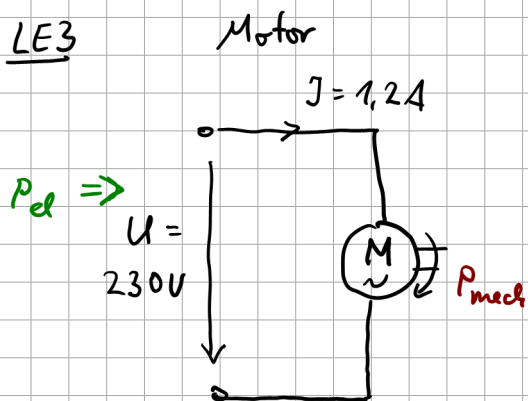
$$\begin{aligned}\underline{S} &= 18,6VA \cdot e^{j104^\circ} \\ &= 18,6VA \cdot [\cos(104^\circ) + j \sin(104^\circ)] \\ &= \underbrace{-4,5W}_P + j \underbrace{18,04var}_Q\end{aligned}$$

$$P < 0 \Rightarrow \text{Erzeuger}$$

$$Q > 0 \Rightarrow \text{induktiv}$$

$$\lambda = \left| \frac{P}{S} \right| = \left| \frac{-4,5W}{18,6VA} \right| = 0,24$$

$$\varphi = \varphi_u - \varphi_i = 28^\circ - (-76^\circ) = +104^\circ$$

LE3

$$\cos \varphi = 0,8$$

$$\text{Wirkungsfaktor } \eta = 0,83$$

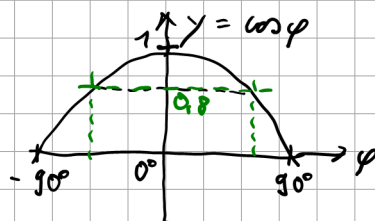
$$P_{\text{mech}} = \eta \cdot P_{\text{el}}$$

a) $P, Q, S, \varphi, \lambda$

$$S = U \cdot I = 230\text{ V} \cdot 1,2\text{ A} = 276\text{ VA}$$

$$P = U \cdot I \cdot \cos \varphi = 276\text{ VA} \cdot 0,8 = 220,8\text{ W}$$

$$\varphi = \arccos 0,8 = 36,9^\circ$$



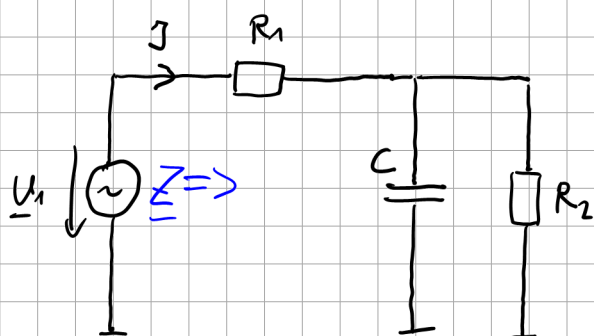
$\Rightarrow$ Motor induktiv $\Rightarrow \varphi > 0$

$$Q = U \cdot I \cdot \sin \varphi = 276\text{ VA} \cdot \sin(36,9^\circ) = 165,7\text{ var}$$

$$\lambda = \cos \varphi = 0,8$$

b) mechanische Leistung

$$\begin{aligned} P_{\text{mech}} &= \eta \cdot P_{\text{el}} \\ &= 0,83 \cdot 220,8\text{ W} \\ &= 183,3\text{ W} \end{aligned}$$

LE 4geg.: $R_1, C, R_2, \omega, \underline{U}_1$ gesucht: $\underline{S}$ von Quelle

Weg: $\underline{S} = \underline{U}_1 \cdot \underline{I}^*$

$$\underline{I} = \frac{\underline{U}_1}{\underline{Z}_{ges}}$$

$$\underline{S} = \underline{U}_1 \cdot \left(\frac{\underline{U}_1}{\underline{Z}_{ges}} \right)^*$$

$$\left(\frac{a}{b} \right)^* = \frac{a^*}{b^*}$$

$$= \underline{U}_1 \cdot \frac{\underline{U}_1^*}{\underline{Z}_{ges}^*}$$

$$= |\underline{U}_1|^2 \cdot \frac{1}{\underline{Z}_{ges}^*}$$

$$a \cdot a^* = |a|^2$$

$$= \frac{U_1^2}{\underline{Z}_{ges}^*}$$

$$\underline{Z} = R_1 + (\underline{Z}_C \parallel R_2) = R_1 + \frac{\frac{1}{j\omega C} \cdot R_2}{\frac{1}{j\omega C} + R_2} = R_1 + \frac{\frac{1}{j\omega C} \cdot R_2}{\frac{1}{j\omega C} + R_2} \cdot \frac{j\omega C}{j\omega C}$$

$$= R_1 + \frac{R_2}{1 + j\omega R_2 C}$$

$$= \frac{R_1(1 + j\omega R_2 C) + R_2}{1 + j\omega R_2 C}$$

$$\underline{Z} = \frac{R_1 + R_2 + j\omega R_1 R_2 C}{1 + j\omega R_2 C}$$

$$\left(\frac{a + jb}{c + jd} \right)^* = \frac{(a + jb)^*}{(c + jd)^*} = \frac{a - jb}{c - jd}$$

$$\underline{Z}^* = \frac{R_1 + R_2 - j\omega R_1 R_2 C}{1 - j\omega R_2 C}$$

$$\underline{S} = \frac{|\underline{U}_1|^2}{\frac{R_1 + R_2 - j\omega R_1 R_2 C}{1 - j\omega R_2 C}}$$

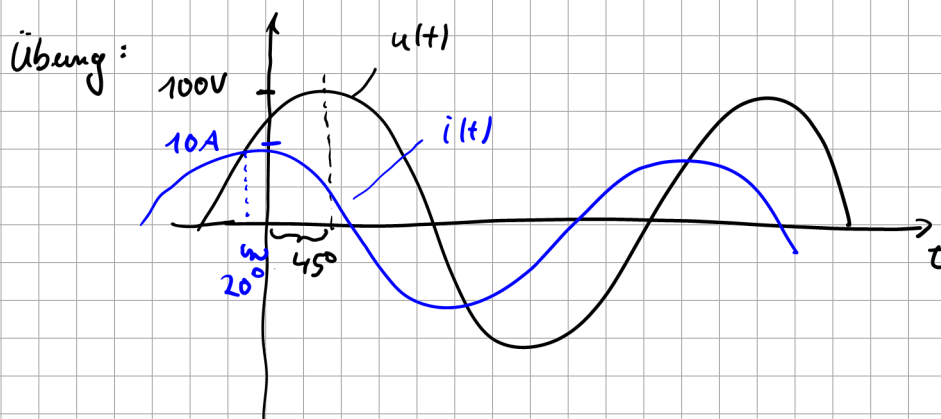
$$\underline{S} = |\underline{U}_1|^2 \cdot \frac{1 - j\omega R_2 C}{R_1 + R_2 - j\omega R_1 R_2 C} \cdot \frac{R_1 + R_2 + j\omega R_1 R_2 C}{R_1 + R_2 + j\omega R_1 R_2 C}$$

$$= U_1^2 \cdot \frac{R_1 + R_2 - \cancel{j\omega R_2 R_1 C} - j\omega R_2^2 C + \cancel{j\omega R_1 R_2 C} + \omega^2 R_1 R_2^2 C^2}{(R_1 + R_2)^2 + (\omega R_1 R_2 C)^2}$$

$$\underline{S} = U_1^2 \left[\frac{R_1 + R_2 + \omega^2 R_1 R_2^2 C^2}{(R_1 + R_2)^2 + (\omega R_1 R_2 C)^2} - j \frac{\omega R_2^2 C}{(R_1 + R_2)^2 + (\omega R_1 R_2 C)^2} \right]$$

$P = \operatorname{Re}\{\underline{S}\}$
 $Q = \operatorname{Im}\{\underline{S}\}$

hier ist $Q < 0 \Rightarrow$ kapazitiv



$$\underline{U} = \frac{100V}{\sqrt{2}} \cdot e^{-j45^\circ}$$

$$\underline{I} = \frac{10A}{\sqrt{2}} \cdot e^{+j20^\circ}$$

$$P, Q, S = ?$$

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